DIGITAL PATHOLOGY • DISTRIBUTED CLOUD COMPUTING • MACHINE LEARNING

SPE3D: A Distributed and Scalable 2D/3D Digital Pathology Platform

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Abstract

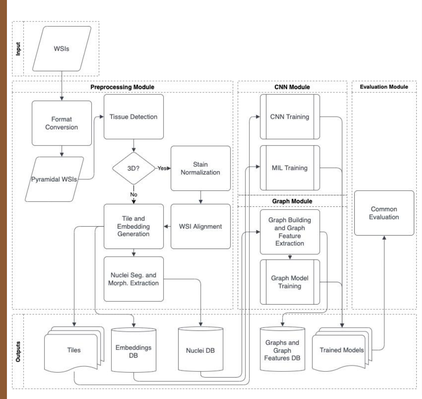
Recent advancements in whole slide digital scanning and increased focus in personalized precision medicine has paved way for research in digital pathology, especially in the application of machine learning (ML) to infer pathological diagnoses. While interest in ML applications in digital pathology is strong, conducting meaningful research in this field demands significant allocation of computational resources. Therefore, there is an urgent need for a scalable digital pathology platform that not only provides tools for Whole Slide Image (WSI) processing but can also efficiently allocate computing resources and upload results to a centralized database. To meet this need, we present SPE3D (Scalable digital Pathology Experimentation in 3D), a state-of-the art, robust, cloud-based digital pathology platform that allows end-to-end processing of pathology images to both train models and run inference at scale. The platform is designed to ingest 2D and 3D tissue samples, process them to create any necessary datasets, and then use the datasets to train and evaluate a collection of models from classical ML models to deep neural networks. The platform runs entirely on a cloud-based infrastructure comprised of CPU and GPU nodes, high-performance machine storage, as well as distributed object storage. The platform can scale the number of nodes required to perform each of the operations based on workload by making extensive use of containerization and workflow-based approaches. Furthermore, the platform supports modern machine learning toolkits, allowing reusability of individual components as well as rapid deployment, prototyping and validation of novel algorithms. Finally, we benchmarked the platform on several 2D and 3D digital pathology datasets to demonstrate its computational capacity as well as the quality of its output datasets.

Background

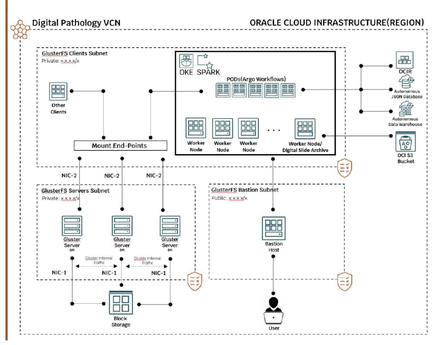
- Novel, high-performing machine learning models are already in use and have shown promising impact on cancer research and clinical care. Application of these models can lead to new clinical workflows and methods that will greatly improve the quality of healthcare (Hong and Fenyö, 2022).
- A Whole Slide Image (WSI) is meant to emulate views through an optical microscope. Thus, the image is taken at very high resolutions, typically producing a file that is several gigabytes in size.
- Developing any high performing machine learning model requires large-scale datasets. In the specific case of digital pathology models, the problem is magnified because each individual datapoint, i.e., a WSI, is so large and requires extensive processing steps even prior to training.
- Cloud computing offers many attractive features for small medium scale organizations: the ability to scale infrastructure up or down as needed, the ability to easily and instantaneously provision and de-provision resources, and reduced IT maintenance costs.
- Integrating distributed computing technologies and deploying on the cloud would ensure a robust platform that can handle the volume of data necessary for AI/ML digital pathology research.

Graphical Abstract

Data Flow



System Architecture



A. Description of SPE3D's Core Modules

Preprocessing Module: transforms raw WSI's into high quality features for analysis and ML

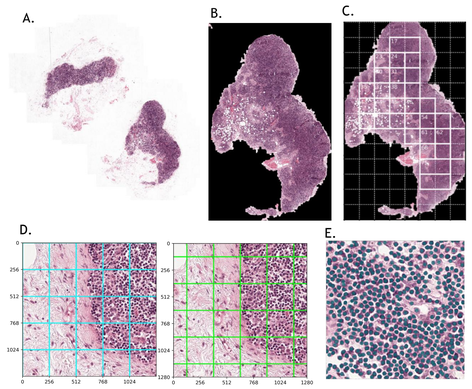


Figure 1. Results from each of the preprocessing submodules. A) A raw WSI from the CAMELYON16 dataset (Bejnordi et al., 2017) is first passed as input. B) The largest piece of tissue is detected and cropped via Gaussian smoothing and Otsu thresholding. C) The cropped tissue is divided into patches of size 4096x4096 for downstream processing and analysis. D) Overlapping tiles and tile embeddings are extracted from patches for nuclei segmentation and model training. E) Nuclei are segmented from tiles using a U-Net (Hou et al. 2020) and mapped back onto the WSI.

CNN Module: trains CNN-based models

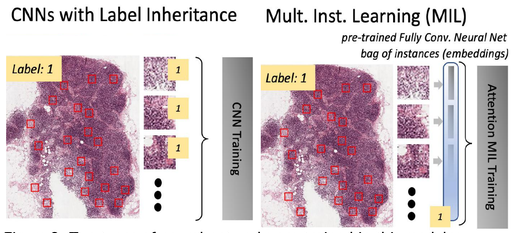


Figure 2: Two types of neural networks are trained in this module: Convolutional Neural Networks and Multiple Instance Learning from ResNet50-generated embeddings (Chuang et al., 2021).

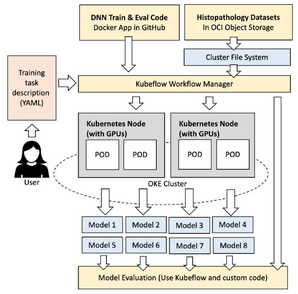


Figure 3: CNN-based models are trained and optimized with hyperparameter search. The module is designed so that all conventional CNN classifiers could be integrated for experimentation. Models are trained in parallel using a cluster of GPU nodes in a distributed architecture. Kubeflow is used to set hyperparameters and view training results.

Graph Module: build nuclei graphs and train graph-based ML models

Definition 1.1 (k-NN Nuclei Graph)**.** A k-Nearest Neighbor (k-NN) nuclei graph is a graph $G = (V, E)$ where V is the set of segmented nuclei and, between an arbitrary pair of vertices (v_a, v_b), E exists iff

1. $D(v_a, v_b) < r$, and
2. the degree of v_a and v_b are both less than k.

where D is a distance metric and r and k are integers.

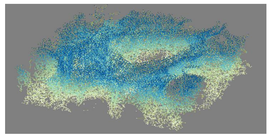


Figure 4. Isometric view of a 3D nuclei graph.

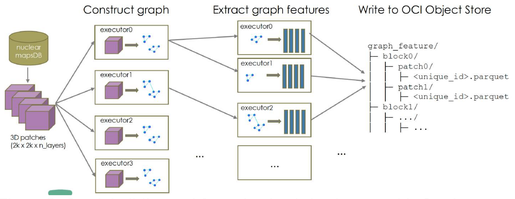


Figure 5. Graph building and featurization is implemented in Spark Spark (Salloum et al., 2016). For graph construction, patches are distributed across executors. For efficient feature extraction, connected subgraphs are distributed across executors.

A. Description of SPE3D's Core Modules (cont'd)

Evaluation Module: fair and unified evaluation of all output ML models

Definition 1.2 (Multi-level Prediction Evaluation)**.** Let $\vec{y}_{predprob_{B_j}}$ be a $1 \times n$ vector of predictions probabilities for patient B_j. When $n > 1$, predictions are on the patch level, whereas when $n = 1$, predictions are on the patient level. Let $\vec{w}_{pred_{B_j}}$ be a $1 \times n$ vector of prediction weights. $\vec{w}_{predprob_{B_j}} = [1]$ for patient level predictions. Thus, the patient level prediction probability is

$$Y_{predprob_{B_j}} = \frac{1}{n} \sum \vec{w}_{predprob_{B_j}} \cdot \vec{y}_{predprob_{B_j}}$$

. Let t be the threshold for prediction, such that the predicted class is 1 if $Y_{predprob_{B_j}} \geq t$ and 0 otherwise. Now, given $\vec{Y}_{predprob}$ of patient level prediction probabilities and $\vec{Y}_{true}$ of ground truth labels, we can calculate the ROC to find optimal threshold t^*. Thus, the optimal class predictions is defined as

$$\vec{Y}_{pred} = \mathbf{1}[\vec{Y}_{predprob} \geq t^*]$$

where $\mathbf{1}$ is the indicator function. With $\vec{Y}_{pred}$ and $\vec{Y}_{true}$, standard ML classification metrics such as accuracy, F1 score, etc can be calculated.

Visualization Module: HistomicsUI Integration

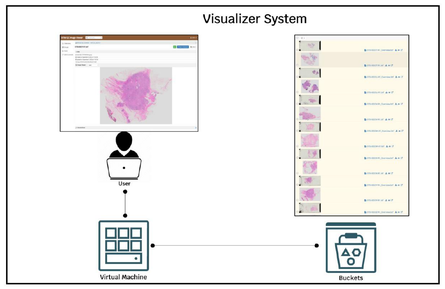


Figure 6. Outputs from each step of modules can be visualized by leveraging Digital Slide Archive and HistomicsUI. The Digital Slide Archive (DSA) is a containerized web-based platform that provides the ability to store, manage, visualize, and annotate large digital pathology imaging data sets. HistomicsUI is part of DSA and is used as an interface to visualize slides and manage annotations. DSA is deployed on one of the nodes of the Kubernetes cluster as a docker container. This system can organize images from a variety of asset stores such as local file systems and OCI Object store. It also provides User access controls.

B. Benchmarking SPE3D

Data Quality

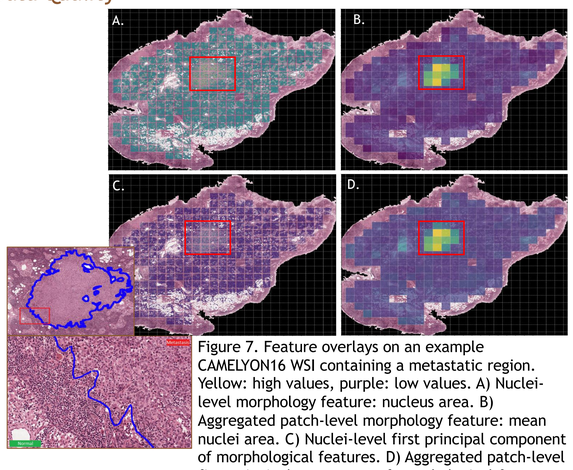


Figure 7. Feature overlays on an example CAMELYON16 WSI containing a metastatic region. Yellow: high values, purple: low values. A) Nuclei-level morphology feature: nucleus area. B) Aggregated patch-level morphology feature: mean nuclei area. C) Nuclei-level first principal component of morphological features. D) Aggregated patch-level first principal component of morphological features.

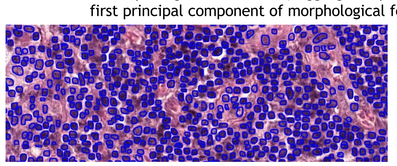


Figure 8. WSI overlay of of U-net nuclei segmentation results.

Acknowledgements

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